Graphical Abstract

Co-Optimization of RES Capacity, Substation Selection, and Cable Sizing for Green Hydrogen LCOH Under Stochastic Electricity Prices in Distribution Networks: A Danish DK1 Case Study

Sreelatha Aihloor Subramanyam, Amjad Anvari-Moghaddam

Highlights

Co-Optimization of RES Capacity, Substation Selection, and Cable Sizing for Green Hydrogen LCOH Under Stochastic Electricity Prices in Distribution Networks: A Danish DK1 Case Study

Sreelatha Aihloor Subramanyam, Amjad Anvari-Moghaddam

- Research highlight 1
- Research highlight 2

Co-Optimization of RES Capacity, Substation Selection, and Cable Sizing for Green Hydrogen LCOH Under Stochastic Electricity Prices in Distribution Networks: A Danish DK1 Case Study

Sreelatha Aihloor Subramanyam

Department of Energy, Aalborg University, Aalborg, 9220, Aalborg, Denmark

Amjad Anvari-Moghaddam

Department of Energy, Aalborg University, Aalborg, 9220, Aalborg, Denmark

Abstract

Abstract text.

Keywords:

1. Introduction

PtX electrolysers at the 20–300 MW scale represent a new class of heavy, price-responsive industrial loads connecting at sub-transmission voltage in Danish distribution grids. Unlike passive loads, they can shift consumption to low or negative DK1 price periods, creating a bidirectional optimization problem: the operator wants to minimize LCOH while the DSO wants to minimize grid expansion cost. The optimal sizing and location of co-located or remote RES determines both outcomes simultaneously. No existing study co-optimizes RES capacity, substation selection, and cable sizing inside a distribution-grid-constrained stochastic model for a real scaling trajectory from 20 MW to 300 MW. Section text. See Subsection 1.1.

1.1. Example Subsection

Subsection text.

Table 1: Sets and Indices

Set	Symbol	Description
RES sites	I	{collocated_FRN, collocated_YRJ, remote$_1$, remote$_2$}
Substations	J	{FRN, YRJ}
Cable types	K	{cable_33kV_A, cable_33kV_B, cable_132kV_A, ...}
Collection cable types	K_1	Cable catalog for RES→PCC corridors (remote sites only), $K_1 \subseteq K$ or a distinct voltage-class catalog
Connection cable types	K_2	Cable catalog for PCC→substation / collocated corridors, $K_2 \subseteq K$
Time periods	T	$\{1, 2, \ldots, 8760\}$ hours (one year, hourly resolution)
Scenarios	Ω	$\{\omega_1, \ldots, \omega_{50}\}$, joint price–RES scenarios

2. Methodology

3. Problem Framework

The problem is formulated as a two-stage stochastic MILP. In the *first stage*, before uncertainty is realized, the planner decides the installed RES capacity, the substation to which each RES site is connected, and the cable type for each corridor. In the *second stage*, after a scenario ω (jointly comprising the DK1 electricity price trajectory and the wind/solar capacity factor trajectory) is revealed, the operator decides the hourly electrolyser dispatch, grid import, and hydrogen storage policy. The stage structure is

$$\underbrace{\{P_w^{\mathrm{RES}}, P_s^{\mathrm{RES}}, x_{ij}, y_{ijk}\}}_{\text{Stage 1: Investment}} \longrightarrow \underbrace{\omega \sim \Omega}_{\text{Uncertainty realized}} \longrightarrow \underbrace{\{p_t^\omega, g_t^\omega, r_t^\omega, q_t^\omega, s_t^\omega\}}_{\text{Stage 2: Operation}}. \quad (1)$$

4. Sets and Indices

Remote sites are further partitioned by the parameter $\delta_i \in \{0, 1\}$, with $\delta_i = 1$ for $i \in I_{\mathrm{remote}}$ and $\delta_i = 0$ for $i \in I_{\mathrm{colloc}}$ (Section 8.4).

Table 2: Investment and Electrolyser Parameters

Symbol	Description	Unit
c_w	Wind capacity investment cost	€/MW
c_s	Solar capacity investment cost	€/MW
c_k^{cable}	Cable cost, type k	€/km
L_{ij}	Distance matrix, site i to substation j	km
r	Discount rate	%
n	Project lifetime	years
CRF	Capital recovery factor, CRF $= \dfrac{r(1+r)^n}{(1+r)^n - 1}$	–
$P_{\text{elec}}^{\min}$	Minimum electrolyser load	MW
$P_{\text{elec}}^{\max}$	Maximum electrolyser capacity	MW
η	Electrolyser efficiency	kg H_2/MWh
$\Delta P^{\max}$	Ramp-rate limit	MW/h
D	Annual H_2 production target	tonnes/yr

5. Parameters

Remark 1 (distance matrix): L_{ij} is a full $|I| \times |J|$ matrix, not a scalar, since the RES-site-to-substation distance differs for every pair (i, j) (e.g., $L_{\text{collocated_FRN,FRN}} = 0$ km, while a remote site may have $L_{ij} \in [10, 50]$ km depending on j).

Remark 2 (equiprobable scenarios): unless fitted otherwise from historical data, $\pi_\omega = 1/|\Omega| = 1/50$ for all ω, with $\sum_{\omega \in \Omega} \pi_\omega = 1$.

6. Decision Variables

By construction, first-stage variables carry no scenario index ω; the non-anticipativity of these variables is stated formally in Section 10.

Table 3: Grid, Storage, and Scenario Parameters

Symbol	Description	Unit
$T_j^{\max}$	Transformer thermal limit, substation j	MVA
$F_k^{\max}$	Cable type k thermal limit	MW
F_{mn}^{fixed}	Existing branch (m,n) thermal limit	MW
R_{ij}, X_{ij}	Branch resistance, reactance	Ω
V_{base}	Nominal voltage	kV
$V^{\min}, V^{\max}$	Voltage bounds	p.u.
$S^{\max}$	Buffer storage capacity	tonnes
d_t	Hourly H_2 offtake demand	tonnes/h
η_{compress}	Compression efficiency	–
π_ω	Probability of scenario ω	–
λ_t^ω	DK1 spot price, period t, scenario ω	€/MWh
$CF_{w,t}^\omega$	Wind capacity factor, period t, scenario ω	–
$CF_{s,t}^\omega$	Solar capacity factor, period t, scenario ω	–

Table 4: First-Stage (Investment) Variables

Variable	Type	Description
P_w^{RES}	$\mathbb{R}_{\geq 0}$	Installed wind capacity [MW]
P_s^{RES}	$\mathbb{R}_{\geq 0}$	Installed solar capacity [MW]
x_{ij}	$\{0,1\}$	1 if RES site i connects to substation j
y_{ijk}	$\{0,1\}$	1 if cable type k is selected for corridor (i,j)

7. Objective Function

7.1. Nonlinearity of the Direct LCOH Objective

A naïve formulation defines the objective as the ratio of the expected annualized cost to the expected annual hydrogen output,

$$\text{LCOH} = \frac{\text{CRF} \cdot \text{CAPEX} + \mathbb{E}_\omega[\sum_t \lambda_t^\omega g_t^\omega \Delta t]}{\mathbb{E}_\omega[\sum_t q_t^\omega]}. \tag{2}$$

Because (2) divides two expectations that are themselves linear functions of the decision variables, the resulting expression is *nonlinear* and cannot be

Table 5: Second-Stage (Operational) Variables, $\forall t \in T,\ \forall \omega \in \Omega$

Variable	Type	Description
p_t^ω	$\mathbb{R}_{\geq 0}$	Electrolyser power consumption [MW]
g_t^ω	$\mathbb{R}_{\geq 0}$	Grid import [MW]
r_t^ω	$\mathbb{R}_{\geq 0}$	RES output delivered to electrolyser [MW]
q_t^ω	$\mathbb{R}_{\geq 0}$	Hydrogen produced [tonnes/h]
s_t^ω	$\mathbb{R}_{\geq 0}$	H_2 buffer storage level [tonnes]
$P_{ij,t}^\omega$	$\mathbb{R}$ (free)	Active power flow, branch (i,j) [MW]
$Q_{ij,t}^\omega$	$\mathbb{R}$ (free)	Reactive power flow, branch (i,j) [MVAr]
$v_{n,t}^\omega$	$\mathbb{R}_{\geq 0}$	Squared voltage magnitude, node n [kV^2]

Table 6: Investment Parameters for the Danish Power-to-X Context

Symbol	**Description**	**Typical Value**	**Unit**	**Danish Context / Data Source**
c_w	Wind capacity investment cost	1.6×10^6	€/MW	Onshore wind turbines in Denmark [12, 16]
c_s	Solar capacity investment cost	0.45×10^6	€/MW	Utility-scale solar PV in Denmark [16]
c_k^{cable}	Cable cost per type k	0.15×10^6	€/km	33 kV AC collector grid cables [10]
L_{ij}	Distance matrix (RES to substation)	[0 to 35]	km	Collocated (0 km) or remote onshore (5–35 km) [10]
r	Discount rate (WACC)	$5.0 - -6.0$	%	Standard Danish PtX financial WACC [17, 12]
n	Project lifetime	20	years	Typical electrolyser facility project life [12, 14]
CRF	Capital recovery factor	0.0872	–	Calculated at $r = 6\%$, $n = 20$ years [10, 12]

handled directly by a MILP solver. This objective is therefore rejected.

7.2. Linear Reformulation (Fixed Production Target)

Let the annual hydrogen production target D be fixed as an exogenous parameter, taken from the offtake contract. The model then minimizes total

Table 7: Electrolyser and Compressor Specifications

Symbol	Description	Typical Value	Unit	Danish Context / Data Source
$P_{\text{elec}}^{\min}$	Minimum electrolyser load	2.4	MW	20% of nominal load for alkaline systems [17]
$P_{\text{elec}}^{\max}$	Maximum electrolyser capacity	20.0	MW	HySynergy Phase 1 target capacity [10, 17]
η	Electrolyser system efficiency	19.0	kg H_2/MWh	Equivalent to $\sim 70\%$ HHV efficiency (52.6 kWh/kg) [17]
$\Delta P^{\max}$	Ramp rate limit	$0.15 P_{\text{elec}}^{\max}$	MW/h	Hourly ramp limits for system stability [9]
D	Annual H_2 production target	870–1740	tonnes/yr	Contractual baseline for Danish off-takers [5]
η_{compress}	Auxiliary compressor energy	1.67	kWh/kg H_2	Mechanical compression from 30 bar to 350 bar [14]

annualized cost,

$$\min_{\substack{P_w^{\text{RES}}, P_s^{\text{RES}}, \\ x_{ij}, y_{ijk}, \\ p_t^\omega, \ldots}} \quad TC = \text{CRF} \cdot \text{CAPEX} + \sum_{\omega \in \Omega} \pi_\omega \sum_{t \in T} \lambda_t^\omega \, g_t^\omega \, \Delta t, \tag{3}$$

and the levelized cost of hydrogen is recovered by post-optimization division by the (fixed, scenario-independent) constant D:

$$\text{LCOH}^* = \frac{TC^*}{D} \quad [/\text{kg H}_2]. \tag{4}$$

Equation (4) is linear in TC^* since D is a constant, resolving the nonlinearity in (2).

7.3. CAPEX Expansion

$$\text{CAPEX} = c_w P_w^{\text{RES}} + c_s P_s^{\text{RES}} + \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} c_k^{\text{cable}} \, L_{ij} \, y_{ijk}. \tag{5}$$

Table 8: Grid and Network Transmission Parameters

Symbol	Description	Typical Value	Unit	Danish Context / Data Source
$T^{\max}$	Transformer thermal limit	46.0	MVA	Standard Danish DSO substation coupling limit [6]
$F_k^{\max}$	Cable type k thermal limit	20–30	MW	Standard 33 kV collector cable thermal capacity [11]
F_{mn}^{fixed}	Existing branch thermal limit	55.0	MW	Existing branch constraint [7]
X_{ij}	Line reactance	0.13	pu	Typical reactance for Danish distribution lines [7]
V_{base}	Nominal base voltage	33.0	kV	Grid collector voltage level [10]
$V^{\min}, V^{\max}$	Voltage limits	[0.95–1.05]	pu	Standard operational voltage bounds [10]
λ_{TSO}	TSO consumption tariff	15.60	€/MWh	West Denmark (DK1) Energinet tariff [14]
λ_{DSO}	DSO network tariff	5.36	€/MWh	West Denmark (DK1) Radius Elnet distribution tariff [14]

Table 9: Hydrogen Storage and Demand Parameters

Symbol	Description	Typical Value	Unit	Danish Context / Data Source
$S^{\max}$	Buffer storage capacity	10.0	tonnes	HySynergy mid-scale tank capacity [10]
d_t	Baseline hourly H_2 demand	0.18	tonnes/h	Standard industrial/transport continuous demand [14]
SOC_{ini}	Initial storage state of charge	0.50	–	Standard initial storage setpoint (50% capacity) [17]

For remote RES sites ($i \in I_{\text{remote}}$), a two-corridor cable model applies: Corridor 1 connects the RES site to a point of common coupling (PCC) and must be sized to the full RES output, while Corridor 2 connects the PCC to the substation and is sized to the net power flow. Because the two corridors

Table 10: Stochastic Scenario and Capacity Factor Parameters

Symbol	Description	Typical Value	Unit	Danish Context / Data Source
π_ω	Scenario probability	0.02	–	Equiprobable 50-scenario tree allocation ($\pi_\omega = 1/50$) [10]
λ_t^ω	DK1 day-ahead spot price	25.02–78.50	€/MWh	Nord Pool historical spot price mean for DK1 zone [16]
$CF_{w,t}^\omega$	Onshore wind capacity factor	0.42	–	Jammerbugt wind profile average capacity factor [15]
$CF_{s,t}^\omega$	Solar capacity factor	0.11–0.15	–	Average Danish PV annual generation profile [10]

serve different functions and may draw from different voltage-class catalogs, cable-type selection is drawn from two distinct sets, K_1 (collection) and K_2 (connection), rather than a single shared set K:

$$\text{CAPEX}_{\text{cable}}^{\text{remote}} = \sum_{i \in I_{\text{remote}}} \sum_{j \in J} \Bigg[\sum_{k_1 \in K_1} c_{k_1}^{\text{collect}} \, L_i^{\text{collect}} \, z_{ijk_1} + \sum_{k_2 \in K_2} c_{k_2}^{\text{connect}} \, L_{ij}^{\text{connect}} \, z_{ijk_2} \Bigg], \tag{6}$$

where $z_{ijk_1} \in \{0,1\}$ selects the Corridor-1 cable type from K_1 and $z_{ijk_2} \in \{0,1\}$ selects the Corridor-2 cable type from K_2. Collocated sites ($i \in I_{\text{colloc}}$) require only Corridor 2, and for these sites the general connection variable y_{ijk} of C2 (Section VI) is understood to range over $k \in K_2$ (i.e. $K \equiv K_2$ in the collocated case, so C1–C2 apply unchanged). For remote sites, $y_{ijk} \equiv z_{ijk_2}$ for $k \in K_2$, and z_{ijk_1} is an additional Corridor-1-only variable activated by $\delta_i = 1$ (C4).

8. First-Stage Constraints

8.1. C1 — Connection Logic

Each RES site connects to at most one substation:

$$\sum_{j \in J} x_{ij} \leq 1 \qquad \forall i \in I. \tag{7}$$

8.2. C2 — Cable Selection Logic

The original constraint $\sum_k y_{ijk} = x_{ij}$ alone is logically loose: when $x_{ij} = 0$, the sum $\sum_k y_{ijk} = 0$ is trivially satisfied, but it does not prevent an individual $y_{ijk} = 1$ paired with another $y_{ijk'} = -1$ being avoided only because variables are binary and non-negative – more importantly it fails to explicitly forbid cable selection when no connection exists in a way that is robust to model extension. The corrected pair of constraints enforces both directions of the logical implication:

$$y_{ijk} \leq x_{ij} \qquad \forall i \in I,\ j \in J,\ k \in K, \tag{8}$$

$$\sum_{k \in K} y_{ijk} = x_{ij} \qquad \forall i \in I,\ j \in J. \tag{9}$$

Equation (8) forbids cable selection absent a connection; (9) forces exactly one cable type when a connection is made.

8.3. C3 — RES Capacity Bounds

$$P_w^{\min} \leq P_w^{\mathrm{RES}} \leq P_w^{\max}, \tag{10}$$

$$P_s^{\min} \leq P_s^{\mathrm{RES}} \leq P_s^{\max}. \tag{11}$$

8.4. C4 — Remote/Collocated Site Indicator

$$\delta_i = \begin{cases} 1, & i \in I_{\mathrm{remote}}, \\ 0, & i \in I_{\mathrm{colloc}}, \end{cases} \tag{12}$$

where δ_i is a fixed binary *parameter* (not a decision variable) that activates the two-corridor CAPEX model of (6).

9. Second-Stage Constraints

All constraints in this section hold $\forall t \in T,\ \forall \omega \in \Omega$ unless otherwise noted.

9.1. C5 — Electrolyser Energy Balance

$$p_t^\omega = r_t^\omega + g_t^\omega. \tag{13}$$

9.2. C6 — RES Availability Bound

$$r_t^\omega \le CF_{w,t}^\omega \, P_w^{\mathrm{RES}} + CF_{s,t}^\omega \, P_s^{\mathrm{RES}}. \tag{14}$$

The capacity factors $CF_{w,t}^\omega$ and the spot price λ_t^ω must be generated *jointly* rather than independently, since the DK1 market exhibits a negative wind–price correlation (high wind output depresses the marginal price). Joint scenario generation follows the method of Papadimitriou *et al.* (2025) [13].

9.3. C7 — Electrolyser Operating Bounds

$$P_{\mathrm{elec}}^{\min} \le p_t^\omega \le P_{\mathrm{elec}}^{\max}. \tag{15}$$

9.4. C8 — Electrolyser Ramp-Rate Limits

$$p_t^\omega - p_{t-1}^\omega \le \Delta P^{\max}, \tag{16}$$

$$p_{t-1}^\omega - p_t^\omega \le \Delta P^{\max}. \tag{17}$$

Equations (16)–(17) are the linearized form of $|p_t^\omega - p_{t-1}^\omega| \le \Delta P^{\max}$.

9.5. C9 — Hydrogen Production

$$q_t^\omega = \eta \, p_t^\omega \, \Delta t. \tag{18}$$

The linear efficiency model is justified at the investment-planning timescale following Baumhof *et al.* (2023) [4]; a nonlinear efficiency curve is reserved for operational dispatch studies outside the scope of this formulation.

9.6. C10 — Annual Production Target

Either a per-scenario production floor,

$$\sum_{t \in T} q_t^{\omega} \geq \frac{D}{|\Omega|} \qquad \forall \omega \in \Omega, \tag{19}$$

or, equivalently in expectation,

$$\sum_{\omega \in \Omega} \pi_{\omega} \sum_{t \in T} q_t^{\omega} \geq D. \tag{20}$$

This constraint is feasibility-coupled to C11 and C13 below; see the consistency remark following C13 and feasibility condition F0 in Section 15.

9.7. C11 — Buffer Storage Dynamics

$$s_{t+1}^{\omega} = s_t^{\omega} + \eta_{\text{compress}}\, q_t^{\omega} - d_t. \tag{21}$$

Compression efficiency η_{compress} is included explicitly; omitting it (as in an earlier draft of this model) implicitly assumes lossless compression, which is not physically justified.

9.8. C12 — Storage Capacity Bounds

$$0 \leq s_t^{\omega} \leq S^{\max}. \tag{22}$$

9.9. C13 — Periodic Storage Boundary

$$s_0^{\omega} = s_T^{\omega} \qquad \forall \omega \in \Omega. \tag{23}$$

Without (23), the optimizer can drain the buffer to zero at the end of the horizon at no cost, which understates LCOH by treating a one-time inventory drawdown as if it were free.

Remark (C10–C13 feasibility coupling — critical). Telescoping the storage recursion (21) over $t = 0, \ldots, T-1$ gives

$$s_T^{\omega} - s_0^{\omega} = \eta_{\text{compress}} \sum_{t \in T} q_t^{\omega} - \sum_{t \in T} d_t. \tag{24}$$

Imposing the cyclic condition (23) therefore forces, for *every* scenario ω,

$$\eta_{\text{compress}} \sum_{t \in T} q_t^{\omega} = \sum_{t \in T} d_t, \tag{25}$$

i.e. total annual hydrogen production is rigidly pinned to the hourly demand profile d_t and is *independent of* ω *and of the target* D *in C10.* If D in (19)–(20) is set independently of d_t, the model is infeasible in *every* scenario, not merely adverse ones, since (19)/(20) and (25) cannot both hold unless $\sum_t d_t/\eta_{\text{compress}} \geq D/|\Omega|$ (resp. $\geq D$) with equality effectively required by (25). This is resolved by feasibility condition F0 (Section 15), which *derives* d_t from D rather than treating them as independent parameters. Under F0, C10 becomes redundant given C11 and C13 and is retained only as a numerical safety check. To absorb rounding-level mismatches from scenario generation, C13 may optionally be relaxed to a tolerance band,

$$-\epsilon\, S^{\max} \;\leq\; s_T^{\omega} - s_0^{\omega} \;\leq\; \epsilon\, S^{\max} \qquad \forall \omega \in \Omega, \tag{26}$$

with $\epsilon \ll 1$ (e.g. $\epsilon = 0.01$), in place of the strict equality (23).

9.10. C14 — Linearized DistFlow Branch Equations

A simple nodal power balance, $f_{ij,t}^{\omega} = \sum_{k \in \text{downstream}(j)} (p_{k,t}^{\omega} - r_{k,t}^{\omega})$, does not model voltage and is rejected. The linearized DistFlow model of Baran and Wu [3] is used instead. Active power balance at node j:

$$P_{ij,t}^{\omega} = \sum_{k:(j,k) \in E} P_{jk,t}^{\omega} + p_{j,t}^{\omega} - r_{j,t}^{\omega} \qquad \forall (i,j) \in E. \tag{27}$$

Reactive power balance at node j:

$$Q_{ij,t}^{\omega} = \sum_{k:(j,k) \in E} Q_{jk,t}^{\omega} + q_{\text{load},j,t}^{\omega} \qquad \forall (i,j) \in E. \tag{28}$$

Voltage-drop equation:

$$v_{j,t}^{\omega} = v_{i,t}^{\omega} - \frac{2\left(R_{ij} P_{ij,t}^{\omega} + X_{ij} Q_{ij,t}^{\omega}\right)}{V_{\text{base}}^2} \qquad \forall (i,j) \in E. \tag{29}$$

Equation (29) is the element omitted from the original nodal formulation; without it, the voltage bounds C15 below have no supporting physics and are vacuous.

Remark (linearization error). Equation (29) omits the quadratic loss term $(R_{ij}^2 + X_{ij}^2)(P_{ij,t}^{\omega\,2} + Q_{ij,t}^{\omega\,2})/V_{\text{base}}^4$ present in the exact (nonlinear) DistFlow model. This is standard practice at the investment-planning timescale and introduces negligible error for lightly loaded MV/HV corridors typical of this application (loading $\lesssim 60\%$ of thermal rating), but the approximation error grows with loading and should be validated post-optimization by substituting the solved flows into the full nonlinear DistFlow equations and checking the residual voltage deviation against $V^{\min}, V^{\max}$ margins.

9.11. C15 — Voltage Bounds

$$(V^{\min})^2 \le v_{n,t}^{\omega} \le (V^{\max})^2 \qquad \forall n \in N, \tag{30}$$

with typical values $V^{\min} = 0.95$ p.u., $V^{\max} = 1.05$ p.u., i.e. $(0.95\,V_{\text{base}})^2 \le v_{n,t}^{\omega} \le (1.05\,V_{\text{base}})^2$.

9.12. C16 — Thermal Limits, New Cable Corridors

$$-F_k^{\max}\, y_{ijk} \le P_{ij,t}^{\omega} \le F_k^{\max}\, y_{ijk} \qquad \forall i \in I,\ j \in J,\ k \in K. \tag{31}$$

9.13. C17 — Thermal Limits, Existing Branches

Existing infrastructure carries a fixed rating that is *independent* of the investment variable x_{ij}; coupling it to x_{ij}, as in an earlier draft, incorrectly zeroes out the existing-branch limit whenever $x_{ij} = 0$:

$$-F_{mn}^{\text{fixed}} \le P_{mn,t}^{\omega} \le F_{mn}^{\text{fixed}} \qquad \forall (m,n) \in E_{\text{existing}}. \tag{32}$$

9.14. C18 — Transformer Thermal Limit

$$\sum_{(m,j)\in E:\, j\in J} P_{mj,t}^{\omega} \le T_j^{\max} \qquad \forall j \in J. \tag{33}$$

10. Non-Anticipativity Constraints

First-stage variables must take a single, scenario-independent value, formalizing the requirement that investment decisions cannot depend on information (the realized scenario) that is not yet available at the time of

investment:

$$P_w^{\mathrm{RES}}(\omega) = P_w^{\mathrm{RES}} \qquad \forall \omega \in \Omega, \tag{34}$$

$$P_s^{\mathrm{RES}}(\omega) = P_s^{\mathrm{RES}} \qquad \forall \omega \in \Omega, \tag{35}$$

$$x_{ij}(\omega) = x_{ij} \qquad \forall \omega \in \Omega, \tag{36}$$

$$y_{ijk}(\omega) = y_{ijk} \qquad \forall \omega \in \Omega. \tag{37}$$

Second-stage variables are indexed by ω throughout and adapt freely to each realized scenario.

11. Complete Model Summary

$$\begin{aligned} \min \quad & TC = \mathrm{CRF}\Big[c_w P_w^{\mathrm{RES}} + c_s P_s^{\mathrm{RES}} + \sum_{i,j,k} c_k^{\mathrm{cable}} L_{ij} y_{ijk}\Big] \\ & \quad + \sum_{\omega \in \Omega} \pi_\omega \sum_{t \in T} \lambda_t^\omega \, g_t^\omega \, \Delta t \end{aligned} \tag{38}$$

s.t. [First stage]

C1–C3: Eqs. (7)–(11)

C2 (corrected): Eqs. (8)–(9)

[Second stage, $\forall t \in T,\ \forall \omega \in \Omega$]

C5–C10: Eqs. (13)–(20)

C11–C13: Eqs. (21)–(23)

[DistFlow grid]

C14–C18: Eqs. (27)–(33)

[Non-anticipativity] Eqs. (34)–(37)

[Variable domains]

$P_w^{\mathrm{RES}}, P_s^{\mathrm{RES}} \geq 0$, continuous

$x_{ij}, y_{ijk} \in \{0, 1\}$, binary

$p_t^\omega, g_t^\omega, r_t^\omega, q_t^\omega, s_t^\omega \geq 0$, continuous

$P_{ij,t}^\omega, Q_{ij,t}^\omega$ free, continuous

$v_{n,t}^\omega \geq 0$, continuous

Post-optimization:

$$\mathrm{LCOH}^* = TC^*/D \quad [/\mathrm{kg\ H_2}], \tag{39}$$

$$\mathrm{VSS} = \mathrm{LCOH}_{\mathrm{det}} - \mathrm{LCOH}_{\mathrm{stoch}}. \tag{40}$$

12. Break-Even Cable Distance Analysis

The break-even distance between remote and collocated RES siting is obtained by parametrically re-solving the full MILP for a sweep of remote distances:

$$L_{\mathrm{remote}} \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}\ \mathrm{km}, \tag{41}$$

fixing $i =$ remote site and $L_{ij} = L_{\mathrm{remote}}$ in each run to obtain $\mathrm{LCOH}_{\mathrm{remote}}(L_{\mathrm{remote}})$, and separately fixing $i \in \{\text{collocated_FRN, collocated_YRJ}\}$ to obtain $\mathrm{LCOH}_{\mathrm{colloc}}$. The break-even distance L^* satisfies

$$\mathrm{LCOH}_{\mathrm{remote}}(L^*) = \mathrm{LCOH}_{\mathrm{colloc}}. \tag{42}$$

13. Value of the Stochastic Solution

1. Solve the full stochastic model (38) to obtain $\mathrm{LCOH}^*_{\mathrm{stoch}}$.
2. Construct expected-value parameters,

$$\lambda_t^{\mathrm{EV}} = \sum_{\omega \in \Omega} \pi_\omega\, \lambda_t^\omega, \tag{43}$$

$$CF_t^{\mathrm{EV}} = \sum_{\omega \in \Omega} \pi_\omega\, CF_t^\omega, \tag{44}$$

   and solve the corresponding single-scenario deterministic model to obtain $\mathrm{LCOH}^*_{\mathrm{det}}$.
3. Compute

$$\mathrm{VSS} = \mathrm{LCOH}^*_{\mathrm{det}} - \mathrm{LCOH}^*_{\mathrm{stoch}}. \tag{45}$$

   $\mathrm{VSS} > 0$ indicates the stochastic formulation yields a materially lower cost than a deterministic average-value approximation and justifies the added modeling complexity; $\mathrm{VSS} \approx 0$ indicates a deterministic model would suffice.

Additionally, the value of storage is quantified by re-solving with $S^{\max} = 0$:

$$\mathrm{VoStorage} = \mathrm{LCOH}_{\text{no storage}} - \mathrm{LCOH}_{\text{10t storage}}. \tag{46}$$

14. Scenario Generation

Historical DK1 hourly spot prices are obtained from Energinet / Center Denmark (2021–2024); wind generation from Energinet municipality-level data; solar generation from ERA5 reanalysis for the Fredericia coordinates. Joint price–capacity-factor scenarios are generated following Papadimitriou *et al.* [13], with the negative wind–price correlation enforced via Cholesky decomposition:

$$\mathrm{Corr}\big(\lambda_t^{\omega},\ CF_{w,t}^{\omega}\big) < 0. \tag{47}$$

An initial pool of 500 Monte Carlo scenarios is reduced to $|\Omega| = 50$ via fast-forward selection [8]; convergence is verified by re-solving at $|\Omega| \in \{20, 30, 50, 100\}$ and confirming LCOH stability.

15. Feasibility Conditions

15.1. F0 — Demand/Target Consistency (Critical, Parameter-Level)

As shown in the Remark following C13 (Section VII), the cyclic storage condition (23) algebraically pins total annual production to the hourly demand profile in every scenario (Eq. (25)). For the model to be well-posed, d_t must therefore be *derived* from the contracted annual target D rather than specified independently. Let $\phi_t \geq 0$, $\sum_{t\in T} \phi_t = 1$, denote a normalized offtake shape (e.g. $\phi_t = 1/|T|$ for flat offtake, or a seasonal/weekly profile supplied by the offtake contract). Define

$$d_t \ := \ \eta_{\text{compress}}\, D\, \phi_t \qquad \forall t \in T, \tag{48}$$

so that $\sum_{t\in T} d_t = \eta_{\text{compress}} D$ holds by construction. This construction reflects standard offtake-contract practice: a facility such as HySynergy signs an annual delivery contract specifying a total volume D (tonnes/yr), and the *hourly* delivery schedule d_t is then derived from that annual commitment via an agreed delivery profile ϕ_t (e.g. flat, seasonal, or weekly) — D and d_t are never independently specified in practice, since d_t is contractually *defined by* D, not an independent input. Treating them as two separate free parameters, as an earlier draft of this model did, has no physical counterpart in how offtake contracts are structured, and is the root cause of the infeasibility in Eq. (25). Under (48), Eq. (25) reduces to $\sum_t q_t^{\omega} = D$ for every ω, which is consistent with, and renders redundant (but not incorrect

to retain as a numerical check), the inequality target C10. **This condition must be enforced at the data -preparation stage, before the MILP is solved**; it is not itself a solver constraint, since d_t and D are both parameters. Analysts who require D and d_t to be genuinely independent (e.g. to test offtake-profile sensitivity separately from annual volume) should instead adopt the relaxed cyclic band (26) with ϵ large enough to absorb the resulting gap $|\,\eta_{\text{compress}} D - \sum_t d_t\,| / S^{\max}$, and must verify ϵ against the physical storage capacity via F3 below.

15.2. F1–F5 — Capacity and Network Feasibility

$$\text{F1 (production):} \quad P_{\text{elec}}^{\max}\,\eta\,|T|\,\Delta t \geq D, \tag{49}$$

$$\text{F2 (grid import):} \quad T_j^{\max} \geq P_{\text{elec}}^{\max} \quad \forall j \in J, \tag{50}$$

$$\text{F3 (storage):} \quad S^{\max} \geq d_t\,\Delta t \quad \forall t \in T, \tag{51}$$

$$\text{F4 (voltage):} \quad (V^{\min})^2 \leq v_{\text{substation}} \leq (V^{\max})^2, \tag{52}$$

$$\text{F5 (ampacity):} \quad \exists k \in K_2 :\ F_k^{\max} \geq P_{\text{elec}}^{\max}. \tag{53}$$

F0 must be checked *first*, since a violation of F0 causes infeasibility in every scenario regardless of F1–F5. Violation of any of F1–F5 by the input data likewise implies model infeasibility independent of the solver and should be checked prior to optimization. Note F3 is now stated $\forall t \in T$ (an earlier version checked it only implicitly); a storage buffer sized only for the average hourly offtake can still violate F3 during peak-demand hours if d_t is non-uniform.

16. Mapping of Constraints to Research Sub-Questions

17. Implementation Notes

- **Solver:** Gurobi or CPLEX (commercial); CBC for testing.
- **Language:** Python with Pyomo or gurobipy.
- **Problem size** (300 MW case, $|\Omega| = 50$): binary variables $\approx |I| \times |J| \times |K| = 4 \times 2 \times 5 = 40$; continuous variables $\approx |T| \times |\Omega| \times (5 + 2|E|) \approx 8760 \times 50 \times 15 \approx 6.5 \times 10^6$; constraint count of comparable order.
- **Expected solve time:** 30 min–2 h, depending on network size.

Table 11: Constraint-to-Sub-Question Mapping

Constraint(s)	Sub-question addressed
C1, C2, C3	SQ1 — optimal RES sizing and substation location
C14c (voltage drop) + C15	SQ2 — LCOH sensitivity to grid connection point
Objective ($\pi_\omega, \lambda_t^\omega$)	SQ3 — stochastic DK1 price effect on investment
C14, C16, C17, C18, C8	SQ4 — grid impact: loading, voltage, ramp rate
Two-corridor CAPEX, parametric L	SQ5 — break-even cable distance
C11, C12, C13, F0	SQ6 — storage adequacy and VSS

Linearization summary (absolute-value constraints C8 and C16):

$$p_t^\omega - p_{t-1}^\omega \le \Delta P^{\max}, \qquad p_{t-1}^\omega - p_t^\omega \le \Delta P^{\max}, \tag{54}$$

$$P_{ij,t}^\omega \le F_k^{\max} y_{ijk}, \qquad -P_{ij,t}^\omega \le F_k^{\max} y_{ijk}. \tag{55}$$

Appendix A. Example Appendix Section

Appendix text.
Example citation (remove this line in the final manuscript).

References

[1] Leslie Lamport, *LaTeX: a document preparation system*, Addison Wesley, Massachusetts, 2nd edition, 1994.

[2] Reference details to be added for <label>.

[3] Reference details to be added for BaranWu1989.

[4] Reference details to be added for Baumhof2023.

[5] Reference details to be added for Christensen2024.

[6] Reference details to be added for CoOptimization2022.

[7] Reference details to be added for ElectricityMarketBidding2026.

[8] Reference details to be added for HeitschRomisch2003.

[9] Reference details to be added for Johnsen2025.

[10] Reference details to be added for MathematicalFormulation.

[11] Reference details to be added for Nami2025.

[12] Reference details to be added for Osman2026.

[13] Reference details to be added for Papadimitriou2025.

[14] Reference details to be added for Saretta2023.

[15] Reference details to be added for SogaardDeakin2024.

[16] Reference details to be added for Sorrenti2023.

[17] Reference details to be added for Zheng2023.